

HUADIAN, China

WMO: 542730

Lat: 42.9783N

Lon: 126.7598E

Elev: 264

StdP: 98.19

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.7	(c) -27.9	(d) -33.9	(e) 0.2	(f) -30.5	(g) -31.1	(h) 0.2	(i) -27.6	(j) 9.3	(k) -11.9	(l) 7.8	(m) -12.2	(n) 0.3	(o) 70	(p) 0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	30.8	22.0	29.5	21.7	28.3	21.1	24.6	28.5	23.7	27.4	22.8	26.3	3.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.8	27.1	22.5	17.8	26.1	21.7	16.8	25.1	76.3	28.6	72.7	27.5	69.2	26.4	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.4	(b) 7.2	(c) 6.2	(e) -33.6	(f) 33.0	(g) 2.9	(h) 1.3	(i) -35.7	(j) 33.9	(k) -37.4	(l) 34.7	(m) -39.0	(n) 35.5	(o) -41.1	(p) 36.4		
(a) 8.4	(b) 7.2	(c) 6.2	(e) -33.7	(f) 25.6	(g) 2.9	(h) 1.1	(i) -35.8	(j) 26.4	(k) -37.5	(l) 27.0	(m) -39.1	(n) 27.7	(o) -41.2	(p) 28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-16.9	-11.4	-1.9	7.4	14.9	20.5	23.2	21.9	15.8	7.2	-3.2	-13.4
	DBStd	14.50	4.94	5.91	5.57	4.57	3.65	2.87	2.26	2.74	3.42	4.40	6.02	5.59
	HDD10.0	3141	833	599	368	103	4	0	0	0	2	109	396	726
	HDD18.3	5077	1091	833	626	327	117	13	1	5	90	345	645	984
	CDD10.0	1471	0	0	0	26	156	314	409	368	175	22	0	0
	CDD18.3	367	0	0	0	1	11	76	152	114	13	0	0	0
	CDH23.3	2718	0	0	0	21	182	609	1082	721	99	4	0	0
	CDH26.7	689	0	0	0	2	41	167	301	170	8	0	0	0
Wind	WSAvg	2.2	1.8	2.1	2.8	3.0	2.8	2.2	1.9	1.7	1.6	2.0	2.1	1.9
Precipitation	PrecAvg	743	9	10	16	43	66	108	188	159	71	39	22	11
	PrecMax	1050	25	29	38	84	115	207	338	291	148	86	59	28
	PrecMin	531	1	1	0	13	18	39	84	42	19	5	6	1
	PrecStd	124	6	7	10	17	27	41	61	77	37	24	14	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.3	6.2	16.0	26.5	30.4	32.2	33.0	31.5	27.8	23.2	14.7	3.0
		MCWB	-1.7	1.5	8.3	13.3	17.1	19.3	24.7	24.2	20.0	14.2	8.5	1.1
	2%	DB	-3.2	3.3	12.0	22.5	27.7	30.1	30.9	29.9	26.0	20.2	10.7	0.6
		MCWB	-5.1	0.0	5.1	11.7	15.4	19.0	23.6	23.4	18.7	12.5	6.3	-1.3
	5%	DB	-5.8	1.2	9.1	19.8	25.3	28.6	29.6	28.6	24.2	17.7	8.0	-1.9
		MCWB	-7.4	-1.4	3.5	10.2	14.8	19.0	22.9	22.6	17.4	11.1	4.8	-3.6
	10%	DB	-8.3	-1.0	6.7	17.0	23.1	27.0	28.4	27.2	22.7	15.2	5.3	-4.4
		MCWB	-9.6	-3.2	2.2	8.8	13.9	18.9	22.5	21.9	16.4	10.0	2.6	-6.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.7	2.1	8.8	15.3	19.2	22.5	26.1	25.5	21.8	15.2	9.1	1.5
		MCDB	-0.1	5.2	15.4	23.9	27.2	27.5	30.7	29.9	26.0	21.3	13.6	2.5
	2%	WB	-4.9	0.4	5.9	12.5	17.1	21.4	25.0	24.3	20.3	13.7	7.0	-1.3
		MCDB	-3.4	2.9	11.2	20.7	23.9	26.6	29.0	28.2	23.7	18.8	9.9	0.4
	5%	WB	-7.3	-1.3	3.9	10.8	15.8	20.5	24.1	23.6	18.9	12.1	5.1	-3.5
		MCDB	-5.9	1.0	8.7	18.6	22.6	25.6	27.9	27.2	22.5	16.2	7.8	-1.9
	10%	WB	-9.6	-3.3	2.3	9.3	14.7	19.7	23.3	22.7	17.6	10.6	2.7	-5.9
		MCDB	-8.2	-1.0	6.2	16.0	21.4	24.7	26.9	26.0	21.2	14.6	5.2	-4.6
Mean Daily Temperature Range	5% DB	MDBR	13.9	13.7	11.0	12.5	12.1	10.2	8.4	8.8	12.3	12.7	10.4	12.1
		MCDBR	15.5	13.8	14.4	18.3	16.5	13.3	10.6	10.4	14.0	17.0	13.8	13.8
		MCWBR	13.7	10.7	8.4	8.6	6.1	4.7	4.2	4.6	6.9	9.1	9.4	12.1
	5% WB	MCDBR	15.1	13.2	13.9	16.2	13.3	10.4	8.9	8.6	11.2	14.6	12.9	13.6
		MCWBR	13.5	10.5	8.5	8.2	5.6	4.5	4.1	4.2	5.9	8.5	9.1	12.1
Clear-Sky Solar Irradiance	taub	0.364	0.416	0.499	0.590	0.642	0.672	0.632	0.551	0.485	0.521	0.442	0.381	
	taud	1.936	1.831	1.741	1.640	1.590	1.607	1.727	1.902	1.986	1.798	1.885	1.926	
	Ebn at Noon	710	736	723	694	673	655	676	719	735	621	610	641	
	Edh at Noon	123	163	202	242	261	258	226	184	157	167	130	114	
All-Sky Solar Radiation	RadAvg	2.37	3.36	4.29	4.71	5.34	5.32	4.79	4.67	4.30	3.09	2.07	1.84	
	RadStd	0.17	0.18	0.24	0.35	0.47	0.54	0.44	0.41	0.34	0.23	0.23	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page